Migration in the Americas, 1990–2020:

A Network Analysis Approach

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Introduction & Background

There are few social, economic, or political topics as salient today as international migration. Whether due to increased armed conflict and security; economic disparities and opportunities; and/or climate change and crisis, the movement of people has been an increasingly dynamic feature of today's world[1]. As it pertains to the hemisphere of the Americas, immigration is a central theme in the region's history, its present-day predicament, and its future directions[2]. One must only spend a few minutes surfing through what is left of the cable-news channels to be bombarded with diverse, and often competing perspectives. This includes disagreements on what are seemingly basic facts: How many migrants are there? Where to and from where are they traveling? Has there been a change in the rate of migration in the region? Many value judgements have and continue to be made about migrants, and the effects migration has on the economic, political, and even cultural milieu of the countries to which migrants travel, and to a lesser extent, from which they are leaving. There is much still to understand regarding the dynamic reality of migration, and in a world filled with anecdotes, opinions, and judgements, can a rigorous, statistics-guided lens help us sift through the haze?

To this end, I aim to utilize the tools within Network Analysis to explore migration trends in North, Central, and South America over the past three decades. My goal is to develop a clear, yet detailed picture of migration within the region and assess what information—either confirmatory, conflicting, or enlightening—Network Analysis can provide to help illuminate the experiences and realities of millions, both past and present.

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[1] International Organization for Migration. *World Migration Report 2024*. International Organization for Migration, 2024. https://worldmigrationreport.iom.int/msite/wmr-2024-interactive/.

[2] International Organization for Migration. "Latin America and the Caribbean." *World Migration Report 2024*, International Organization for Migration, 2024. https://worldmigrationreport.iom.int/what-we-do/world-migration-report-2024-chapter-3/latin-america-and-caribbean.

Methodology

Access

For this project I accessed data publicly available from the Population Division of the United Nations[3]. This dataset has information about the estimates of the number of international migrants. Most entries are based on national statistics, in most cases obtained from relevant population censuses. Overall, the dataset provides estimates of international migration for the years relevant for the present analysis: 1990, 2000, 2010, 2020.

Management and Cleaning

After downloading the dataset, the countries were filtered based on the following criteria: (1) member of the North, Central, and South American region, (2) have a population of at least 2 million for each of the four periods assessed. The latter was done to isolate those countries likely to be most influential within the network of migration, i.e. countries with sufficiently large populations that are therefore likely to impact migration as either a predominant "target" or "source" of migrants in the region.

Adjacency matrices were created for each of the four time periods, with the values being defined as total number of migrants whose origin and destination nations were the row- and column-nation identifiers, respectively. From this, a "net" adjacency matrix was created, which calculated the net flow of migrants between all possible nation pairs. The matrix was cleaned so that only the positive values were left. Subsequently, a filter for values greater than 10,000 was imposed. Though an admittedly arbitrary number, this was believed to be sufficient to account for notable net-flows of migrants, while also minimizing the probability of overtly dense networks if smaller net-values were considered. From this, a binary matrix was created, with values for net migration flows of greater than 10,000 being converted to a "1", while all others, whether due to insufficient net-migrants or being the pair arrangement that had a net-negative migrant flow, were assigned a "0".

Attributes Added

These matrices were used to create directed networks in R using the igraph package. The subsequent network statistics were calculated, and relevant vertex and edge attributes added:

- **Degree Total**
    - The total degree value for each of the vertices; considers the total number of edges incident on a vertex, regardless of direction. This general score gives an indication of how "active" a country is regarding migration in the region. It doesn't necessarily offer insights into the specific "role" the country plays, which is addressed more appropriately by the subsequent measures.

[3] United Nations, Department of Economic and Social Affairs, Population Division. "International Migrant Stock." *United Nations*, 2020, https://www.un.org/development/desa/pd/content/international-migrant-stock. Accessed 13 Dec. 2024.

- **Degree "OUT"**
    - Number of edges emanating from the vertex of interest. This can provide insight into countries that are "sources" of migrants, i.e. there is net flow out of these countries.

- **Degree "IN"**
    - Number of edges terminating at the vertex of interest. This measure relates to the extent to which a country is a "receiver" of migrants.

- **Hub score:**
    - In the HITS algorithm, hub scores measure the value of a node based on the number of "high-authority" nodes—i.e. nodes that are major targets of edges—it points to. These scores are calculated iteratively, updating each node's hub score to be the sum of the authority scores of the nodes it links to, and then normalizing the scores.

- **Authority Score:**
    - In the HITS algorithm, authority scores measure the value of a node based on the number of high-hub nodes—i.e. nodes that are major sources of edges—that link to it. These scores are calculated iteratively, updating each node's authority score to be the sum of the hub scores of the nodes linking to it, and then normalizing the scores.

- **Edge Attribute: Proportional to Destination Country's Population**
    - An additional step was performed to add greater detail to the edges of the networks. To account for the proportion of migrants flowing between country pairs, the net-migrant value for country pairs was normalized by dividing by the population of the destination country (specifically its population during the assessed year, i.e. 1990; and per 1000 citizens, i.e. dividing by destination country population divided by 1000). These values represent a rate of migration between country pairs, with units being the number of migrants per 1000 citizens in destination country. These rates were then added to the networks as edge attributes. Dividing by the population of receiving country was decided over dividing by the population of the origin country as the analysis aimed to understand migration as a phenomenon of flow to a nation, as opposed to the ramifications of *population drain* on the origin nations.

Exclusion of USA

Because I wanted to punish myself by having to nearly double the length of my entire code…Let's try that again. To address the notable "gravity" that is the United States within the study of migration, additional networks were created, and above-mentioned metrics calculated, that excluded the United States as a node. Though this led to a notable loss in complexity of the network (i.e. cutting the number of edges nearly in half for all four time-periods assessed), it allowed for a more granular assessment of the other nations within the networks and region.

Caveat to the Figures

When generating the figures below, the size and color of the vertices relate to the various vertex-attributes described above. However, for the degree measures, the size and color are relative to all the other nodes in all four networks. This was intentional as it would allow better visual comparison *across* the assessed periods. However, for the networks involving Hub and Authority scores, the size and color of the vertices are relative only to the nodes within the respective network. This is because Hub and Authority scores are normalized within a network such that scores range from 0-1, thus making it meaningless to make vertex attributes (i.e. size and color) relative to nodes across all networks. Therefore, in the comparisons below, there may be some distortion when looking, for example, between the figures for Degree OUT and Hub, which are conceptually similar though the visual networks may be notably different. It is therefore not always useful to compare between figures.

Additionally, see the table below for clarification of the country abbreviations used in the network visualizations.

Table1. Legend for Country Names

Country Name	Abbreviation
Argentina	Arg
Bolivia	Bol
Brazil	Braz
Canada	Can
Colombia	Col
Costa Rica	CR
Ecuador	Ecu
El Salvador	ElSal
Guatemala	Guat
Honduras	Hond
Mexico	Mex
Nicaragua	NG
Panama	Pan
Paraguay	PG
United States	US
Uruguay	UG
Venezuela	VZ

Findings and Discussion

Descriptive

Below is a table of the various networks with their subsequent sizes, orders, densities, and percent change in density from the year prior. The table demonstrates that the order and density increase across time, and at an increasing rate according to the % change in density. Though it may seem interesting to note that the increases in edges across the years are almost entirely accounted for by nation-pairs that do not include the US (e.g. of the 8 new edges between 2000 and 2010, 7 are accounted for by non-US pairs, i.e. "WithOUT US Data" shows an increase in 7 between those respective years), this is merely an artifact of the US having an edge with nearly all other nations across all periods (94-100% of possible edges present for the US), i.e. there is minimal opportunity for the US to increase its degree score.

Table 2. Size and Order of Network Groups

	Order	Size	Density	% change Density (relative to prior year)
With US Data				
1990	43	19	0.126	-
2000	43		0.126	0%
2010	51		0.149	18.3%
2020	64		0.187	25.5%
WithOUT US Data				
1990	26	18	0.085	-
2000	26		0.085	0%
2010	33		0.108	27.1%
2020	46		0.150	38.9%

The rate of migration was also determined from the dataset across the four time periods and normalized by the total population of the region at the given timepoints (Figure 1). Additionally, all migrants were included when calculating this metric (i.e. not limiting by the >10,000 migrant threshold held for the networks). Of note, the rate nearly doubles from 1990 to 2020 (16.2 vs 30.8 per 1000 inhabitants of the region), and otherwise steadily increases across the assessed time periods, in line with current statistics[4,5].

[4] International Organization for Migration. "Latin America and the Caribbean." *World Migration Report 2024*, 2024, https://worldmigrationreport.iom.int/what-we-do/world-migration-report-2024-chapter-3/latin-america-and-caribbean#:~:text=In%202020%2C%20the%20numbers%20of,other%20countries%20in%20the%20region.&text=Source%3A%20UN%20DESA%2C%202021.

[5] United Nations Development Programme. "Crossing Borders: The Unprecedented Growth of Migration within Latin America and the Caribbean." *UNDP*, 2024, https://www.undp.org/latin-america/blog/crossing-borders-unprecedented-growth-migration-within-latin-america-and-caribbean-0#:~:text=Overall%2C%20the%20total%20number%20of,5.3%20million%20to%2011.3%20million.

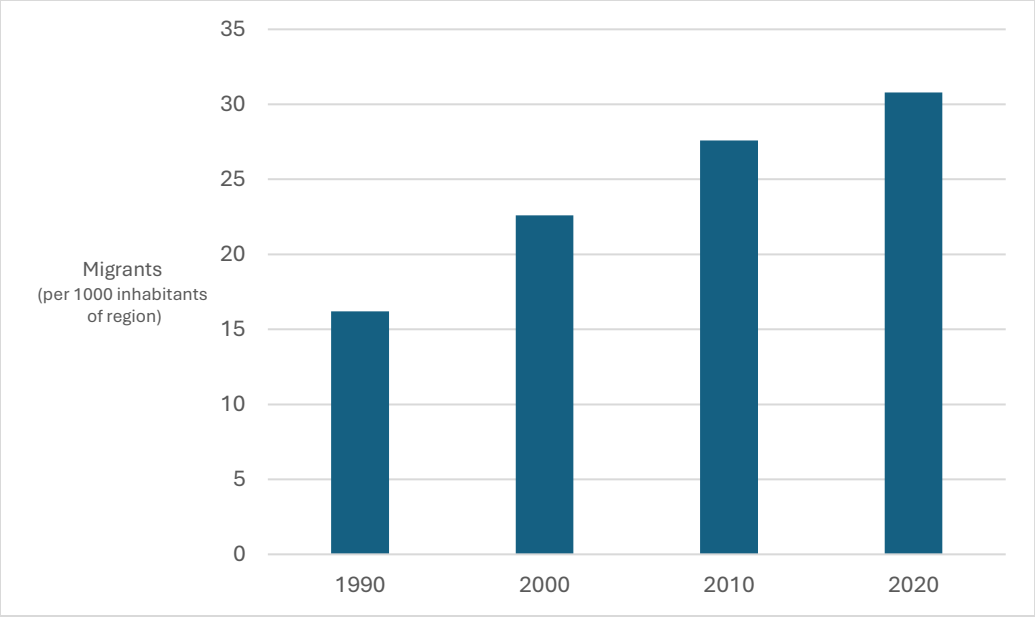

Figure 1. Rate of Migration.

Interpretation

Darien Dynamism

In the various networks presented (figures 2-4), several countries are notable for how their roles appear to change, and even reverse, across the assessed time periods. When considering their geographical distribution, the results provide additional evidence for the migration "hub" that is the Darien Gap, i.e. a narrow, treacherous stretch of minimally inhabited jungle connecting Colombia to Panama, and its neighboring countries[6].

Considering figures 2-4, we see that Venezuela undergoes a seismic shift in its migrant role in the region. During the early years (1990 and 2000) Venezuela was a notable *authority* in the region, a destination for many migrants (figure 3). However, when looking at its greatly diminished vertex size during the year 2020 (figure 3), in conjunction with the dramatic increase in its vertex size in figure 4—denoting its predominance as a *hub*, i.e. a major source of migrants in the region—we see that, over the past 30 years, Venezuela has had the most dynamic migration story within this network of countries.

Colombia also represents a steady *hub* from 2000-2020 (figure 4), though it is particularly interesting to see the edge-reversal between itself and Venezuela between 2010 and 2020 (seen in all figures via the arrow direction), denoting that the net migration in fact flips between the two countries within the decade.

While these countries are converting into senders of migrants, we can see that relevant destination countries include Panama, Ecuador, and even the more distant Chile. The Edges between Venezuela, Colombia, Panama, and Ecuador appear to consolidate into a unidirectional pattern, i.e. the former two countries sending a net flow of migrants to the latter two by 2020. The edge between

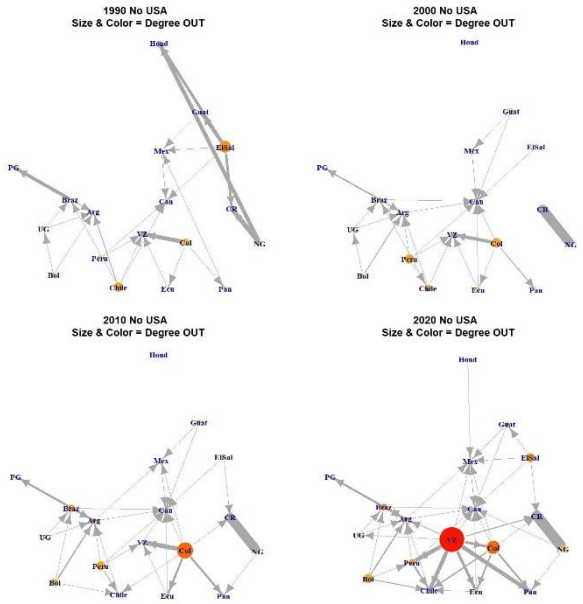

Figure 2. Networks not including the US, with node color and size related to Degree OUT Score.

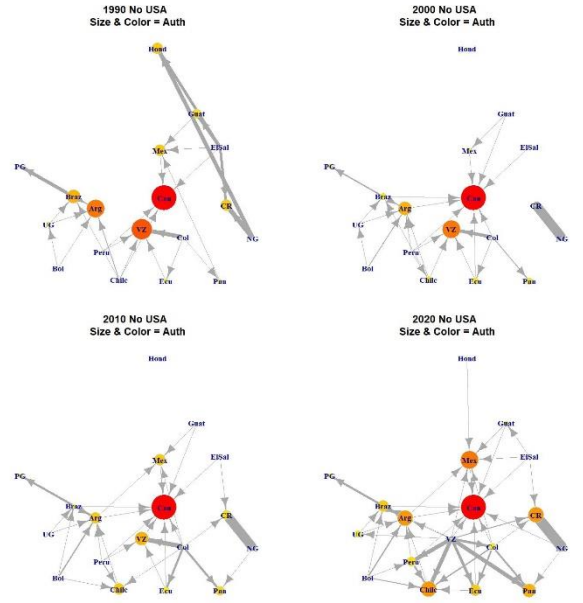

Figure 3. Networks not including the US, with node color and size related to Authority Score.

[6] Center for Strategic and International Studies. "Mind the Darien Gap: A Migration Bottleneck in the Americas." *CSIS*, 2024, https://www.csis.org/analysis/mind-darien-gap-migration-bottleneck-americas.

Venezuela and Ecuador, as well as Venezuela and Panama are notably thick in 2020 when comparing these edges, if present, to prior periods, thus denoting a notable increase in Venezuelan migrants relative to the populations of Ecuador and Panama.

Peru also seems to be impacted by the notable increase in Venezuelan migration in 2020, as its vertex in the *Authority* figure (figure 3) grows slightly in comparison to the three prior periods, while its shared edge with Venezuela switches direction and thickens in 2020, as compared to prior years (figure 3).

Post-Dictatorship Development

Chile is also a country of notable change within the present analysis. It seems to be Venezuela's foil, i.e. role-reversing in a converse fashion. At the turn of the century, Chile is most predominantly a source of migrants. This is seen most clearly in figure 4, with its notably large and deep-red vertex denoting its position as a migrant *hub*, and to some degree in figure 2, though its vertex size and color in this figure is drastically outweighed by that of Venezuela in 2020 (see "Caveats to figures" above for an explanation). But by the years 2010, and especially 2020, Chile becomes a notable authority (figure 3) and destination for migrants, as seen in figure 4 as well, though again here its vertex size is overshadowed by the dominance of Canada as the most predominant destination of migrants. Possible explanations may be the end of the Pinochet dictatorship in 1990, and rapid economic growth, an important consideration for migrants, over the subsequent three decades[7].

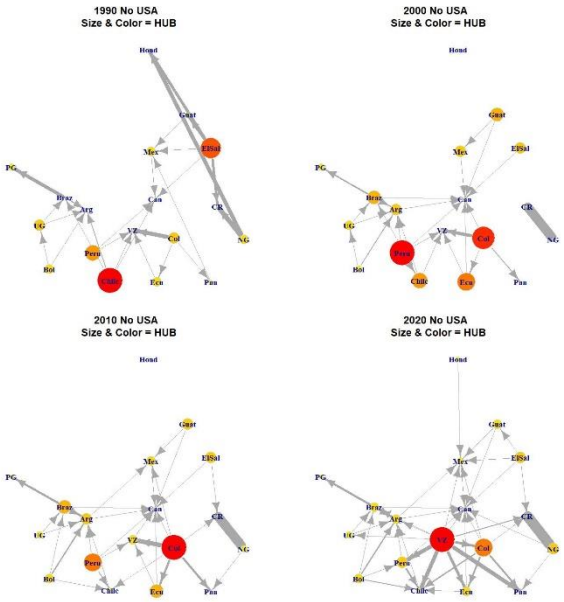

Figure 4. Networks not including the US, with node color and size related to Hub Score.

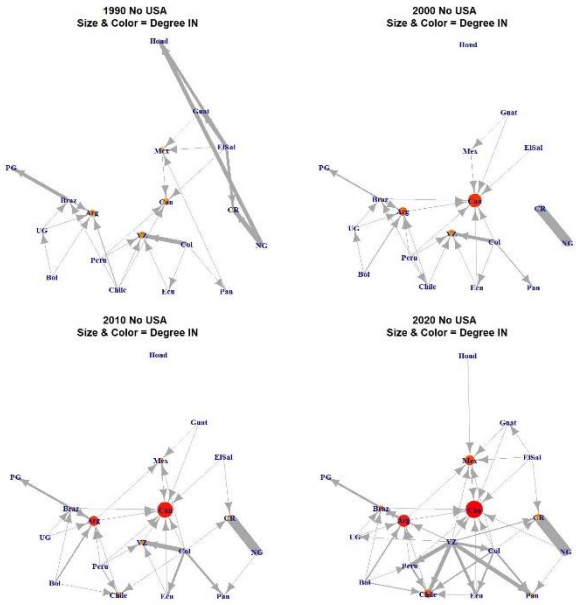

Figure 5. Networks not including the US, with node color and size related to Degree IN Score.

[7] **Stefoni, Carolina, and Dante Contreras.** *Migration in Chile: Trends and Policy Responses in the Period 2000-2021*. United Nations Development Programme, 2022.

United States and Canada:
The "Welcoming" Stars

Quite simply, the US and Canada dominate as major targets for migrants. In figure 6 we see that the US, regardless of time, is the predominant authority, acting as the target of immigration for nearly all represented countries. Canada is initially a smaller player, especially in 1990, but rises to being the second most sought location for migrants by the year 2000. Additionally, apart from Mexico, most edges shared by the US are notably thin, due mainly to its large population size.

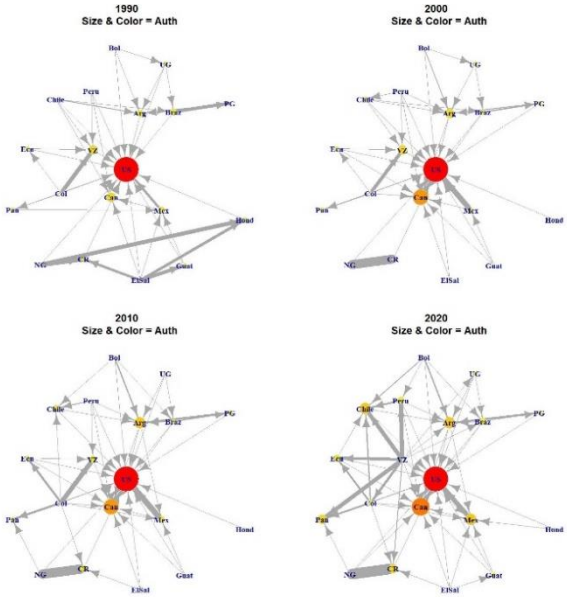

Figure 6. Networks with vertex color and size related to Auth Score.

Conclusions

This early review of migration in the North, Central, and South American region through a Network Analysis lens has reinforced many current conceptions. Firstly, it emphasizes the complexity of migration, not only historically, but in recent years. The drastic growth in migration, linked to increased security concerns, violence, as well as economic and political destabilization, is born out in descriptive statistics from this dataset, particularly the doubling in the migration rate in the region. It also highlights the continued, central role of the United States as a major destination for migrants. The analysis does highlight the complexity and volatility of immigration in the region when excluding the US; immigration is a regional phenomenon affecting all countries beyond just the northern powers. The analysis provides further information about important immigration centers such as the Darien Gap, where geographically-neighboring countries are experiencing notable changes in their network-defined roles in recent years. It also provides additional information for historical trends in migration, especially for changes likely related to issues of civil and political unrest, as was the case in Chile prior to the turn of the century, and more recently in Venezuela with the current Machado regime.

Network Analysis affords one the ability to analyze how various individuals in a network interact. In the case of the American Hemisphere, assessing migration via data-tables of aggregate and net migration, or reviewing aggregated migration rates can be useful for assessing either country specific or region-specific trends. Network Analysis allows us to go a step further by layering together country- and region-specific data. That is to say, the figures presented in, and network-statistics underlying this report demonstrate immigration trends not only for specific countries, but how these micro trends relate to those of other countries, as well as the region at large. The challenge is parsing through these information dense figures and results to find relevant meanings and interpretations therein. To this end, my work here has only begun….

Appendix: This script nearly killed me…

```
rm(list=ls())

#### Code for creating Immigration Matrices based off of UN data #####

#### Relevant links: https://www.un.org/development/desa/pd/content/international-migrant-stock
#####

#### Notes ####

# Data was collected from the UN website. Countries were filtered by those who have a minimum population
of

# 2 million. Parameter for edges is binary with threshold of 10,000 immigrants (i.e. if equal to or greater than

# 10,000 immigrants, assigned a value of 1). The edges also represent net immigration, i.e. subtracting the

# opposing migration pathways between two countries, with the direction of greater size dictating the edge
direction,

# though again, edges are only established if there is a minimum of 10,000 immigrants passing from one
country to the

# other after considering the net migration. Additionally, I added the edge attribute for thickness which is

# associated with this value of net migration divided by the receiving country's population divided by 1000

# (i.e. the units are x immigrants for every 1000 residents).

##########################################################################
#########################################

##########################################################################
#########################################

#setting things up####
```

```
setwd("C:/Users/heblj/OneDrive - Middlebury College/MIIS_Fall 2024/Intro Network Analysis/Final Project/Country Immigration Data Excels")

library(igraph)

load("m_1990.rda")
load("m_2000.rda")
load("m_2010.rda")
load("m_2020.rda")
load("g_90.rda")
load("g_00.rda")
load("g_10.rda")
load("g_20.rda")
load("g_90_NUSA.rda")
load("g_00_NUSA.rda")
load("g_10_NUSA.rda")
load("g_20_NUSA.rda")

layout_preferred_2 <- layout_with_fr(g_20_NUSA)
layout_preferred <- layout_with_fr(g_20)


##########################################################
##########################################################
### WITH THE UNITED STATES INCLUDED ####################
##########################################################
##########################################################


###########################
##### Data from 2020 #####
```

```
##########################

# Network from matrix
network_binary_20 <- read.csv("2020_Immigration Matrix_Binary_simplified.csv",
                header=T,
                row.names=1,
                check.names = F)


#Graphing from Matrix
m_20 <- as.matrix(network_binary_20)

g_20 <- graph_from_adjacency_matrix(m_20,
                        mode = "directed",
                        weighted = NULL)


#Plots to assess
plot(g_20,
    main = "2020_SizeDegree",
    sub = "Vertex by Degree",
    layout = layout_preferred,
    edge.arrow.size = 0.2,
    vertex.size = degree(g_20))


plot(g_20,
    main = "2020_SizeBtwn",
    sub = "Vertex by Btwn",
    edge.arrow.size = 0.2,
    layout = layout_preferred,
```

```
    vertex.size = betweenness(g_20))


##########################
##### Data from 2010 #####
##########################

network_binary_10 <- read.csv("2010_Immigration Matrix_Binary_simplified.csv",
                    header=T,
                    row.names=1,
                    check.names = F)


#Graphing from matrix
m_10 <- as.matrix(network_binary_10)


g_10 <- graph_from_adjacency_matrix(m_10,
                              mode = "directed",
                              weighted = NULL)


plot(g_10,
    main = "2010_SizeDegree",
    edge.arrow.size = 0.2,
    sub = "Vertex by Degree",
```

```
    layout = layout_preferred,
    vertex.size = degree(g_10))


plot(g_10,
    edge.arrow.size = 0.2,
    main = "2010_SizeBtwn",
    sub = "Vertex by Btwn",
    layout = layout_preferred,
    vertex.size = betweenness(g_10))


##########################
##### Data from 2000 #####
##########################

# Network from matrix
network_binary_00 <- read.csv("2000_Immigration Matrix_Binary_simplified.csv",
                    header=T,
                    row.names=1,
                    check.names = F)


#Graphing from matrix
m_00 <- as.matrix(network_binary_00)
```

```
g_00 <- graph_from_adjacency_matrix(m_00,
                                    mode = "directed",
                                    weighted = NULL)


plot(g_00,
     main = "2000_SizeDegree",
     edge.arrow.size = 0.2,
     sub = "Vertex by Degree",
     layout = layout_preferred,
     vertex.size = degree(g_00))


plot(g_00,
     edge.arrow.size = 0.2,
     main = "2000_SizeBtwn",
     sub = "Vertex by Btwn",
     layout = layout_preferred,
     vertex.size = betweenness(g_00))
```

```
##########################
##### Data from 1990 #####
##########################

#Network from Matrix
network_binary_90 <- read.csv("1990_Immigration Matrix_Binary_simplified.csv",
                    header=T,
                    row.names=1,
                   check.names = F)



# #Graph from matrix
 m_90 <- as.matrix(network_binary_90)



 g_90 <- graph_from_adjacency_matrix(m_90,
                              mode = "directed",
                              weighted = NULL)



plot(g_90,
    main = "1990_SizeDegree",
    edge.arrow.size = 0.2,
    sub = "Vertex by Degree",
    layout = layout_preferred,
    vertex.size = degree(g_90, mode = "in"))



plot(g_90,
    main = "2020_SizeBtwn",
```

```
    sub = "Vertex by Btwn",
    edge.arrow.size = 0.2,
    layout = layout_preferred,
    vertex.size = betweenness(g_90))
```

```
##################################################
#######Adding Attributes and modifications #####
##################################################

#Adjusting Country names ####


#Adjusting vertex names so not taking up so much space in network

#for 1990
V(g_90)$name[1] <- "Arg"
V(g_90)$name[2] <- "Bol"
V(g_90)$name[3] <- "Braz"
V(g_90)$name[4] <- "Can"
V(g_90)$name[6] <- "Col"
V(g_90)$name[7] <- "CR"
V(g_90)$name[8] <- "Ecu"
```

```
V(g_90)$name[9] <- "ElSal"
V(g_90)$name[10] <- "Guat"
V(g_90)$name[11] <- "Hond"
V(g_90)$name[12] <- "Mex"
V(g_90)$name[13] <- "NG"
V(g_90)$name[14] <- "Pan"
V(g_90)$name[15] <- "PG"
V(g_90)$name[17] <- "US"
V(g_90)$name[18] <- "UG"
V(g_90)$name[19] <- "VZ"

#for 2000
V(g_00)$name[1] <- "Arg"
V(g_00)$name[2] <- "Bol"
V(g_00)$name[3] <- "Braz"
V(g_00)$name[4] <- "Can"
V(g_00)$name[6] <- "Col"
V(g_00)$name[7] <- "CR"
V(g_00)$name[8] <- "Ecu"
V(g_00)$name[9] <- "ElSal"
V(g_00)$name[10] <- "Guat"
V(g_00)$name[11] <- "Hond"
V(g_00)$name[12] <- "Mex"
V(g_00)$name[13] <- "NG"
V(g_00)$name[14] <- "Pan"
V(g_00)$name[15] <- "PG"
V(g_00)$name[17] <- "US"
V(g_00)$name[18] <- "UG"
V(g_00)$name[19] <- "VZ"

#for 2010
```

```
V(g_10)$name[1] <- "Arg"
V(g_10)$name[2] <- "Bol"
V(g_10)$name[3] <- "Braz"
V(g_10)$name[4] <- "Can"
V(g_10)$name[6] <- "Col"
V(g_10)$name[7] <- "CR"
V(g_10)$name[8] <- "Ecu"
V(g_10)$name[9] <- "ElSal"
V(g_10)$name[10] <- "Guat"
V(g_10)$name[11] <- "Hond"
V(g_10)$name[12] <- "Mex"
V(g_10)$name[13] <- "NG"
V(g_10)$name[14] <- "Pan"
V(g_10)$name[15] <- "PG"
V(g_10)$name[17] <- "US"
V(g_10)$name[18] <- "UG"
V(g_10)$name[19] <- "VZ"

#for 2020
V(g_20)$name[1] <- "Arg"
V(g_20)$name[2] <- "Bol"
V(g_20)$name[3] <- "Braz"
V(g_20)$name[4] <- "Can"
V(g_20)$name[6] <- "Col"
V(g_20)$name[7] <- "CR"
V(g_20)$name[8] <- "Ecu"
V(g_20)$name[9] <- "ElSal"
V(g_20)$name[10] <- "Guat"
V(g_20)$name[11] <- "Hond"
V(g_20)$name[12] <- "Mex"
V(g_20)$name[13] <- "NG"
```

```
V(g_20)$name[14] <- "Pan"
V(g_20)$name[15] <- "PG"
V(g_20)$name[17] <- "US"
V(g_20)$name[18] <- "UG"
V(g_20)$name[19] <- "VZ"
```

```
#Adding Degree as vertex attribute####

###Degree Total ####
# Assuming you have four graphs g1, g2, g3, g4
graphs_td <- list(g_90, g_00, g_10, g_20)
#
## Calculate degree scores for each graph
degree_scores_td <- lapply(graphs_td, degree)


V(g_90)$DegT <- degree_scores_td[[1]]
V(g_00)$DegT <- degree_scores_td[[2]]
V(g_10)$DegT <- degree_scores_td[[3]]
V(g_20)$DegT <- degree_scores_td[[4]]


###Degree IN ####
# Assuming you have four graphs g1, g2, g3, g4
graphs_di <- list(g_90, g_00, g_10, g_20)
#
## Calculate degree scores for each graph
degree_scores_di <- lapply(graphs_di, degree, mode = "in")
```

```
V(g_90)$DegI <- degree_scores_di[[1]]
V(g_00)$DegI <- degree_scores_di[[2]]
V(g_10)$DegI <- degree_scores_di[[3]]
V(g_20)$DegI <- degree_scores_di[[4]]


###Degree OUT ####
# Assuming you have four graphs g1, g2, g3, g4
graphs_do <- list(g_90, g_00, g_10, g_20)
#
## Calculate degree scores for each graph
degree_scores_do <- lapply(graphs_do, degree, mode = "out")


V(g_90)$DegO <- degree_scores_do[[1]]
V(g_00)$DegO <- degree_scores_do[[2]]
V(g_10)$DegO <- degree_scores_do[[3]]
V(g_20)$DegO <- degree_scores_do[[4]]


#Creating Hub and Auth Scores####

######Hubs and Auth for 1990####

#Extract and view the hub and authority scores
hub_g_90 <- hits_scores(g_90)$hub
auth_g_90 <- hits_scores(g_90)$authority
```

```
######Hubs and Auth for 2000####
hub_g_00 <- hits_scores(g_00)$hub
auth_g_00 <- hits_scores(g_00)$authority


######Hubs and Auth for 2010####
hub_g_10 <- hits_scores(g_10)$hub
auth_g_10 <- hits_scores(g_10)$authority


######Hubs and Auth for 2020####
hub_g_20 <- hits_scores(g_20)$hub
auth_g_20 <- hits_scores(g_20)$authority

###Add Hubs and Auth as Vertex Attributes ####

#Add hubs as vertex attribute
##since scores from 0 to 1, need to times 10 to get reasonable value for attribute use
##and add 1 so that the lowest hub and auth score is not 0, which would affect attribute use.
##Also, rounding these values to simplify the colorpalette function below for vertex coloring.
V(g_90)$Hub <- round(hub_g_90*10+1, 0)
V(g_00)$Hub <- round(hub_g_00*10+1, 0)
V(g_10)$Hub <- round(hub_g_10*10+1, 0)
V(g_20)$Hub <- round(hub_g_20*10+1, 0)

#Add auth as vertex attribute
V(g_90)$Auth <- round(auth_g_90*10+1, 0)
V(g_00)$Auth <- round(auth_g_00*10+1, 0)
V(g_10)$Auth <- round(auth_g_10*10+1, 0)
```

```
V(g_20)$Auth <- round(auth_g_20*10+1, 0)


#Create color attributes based on DegT, DegI, DEGO, Hub, Auth####


######Color for DegTotal####
# # Find the maximum degree score for each vertex across all networks
max_degree_value_td <- max(unlist(degree_scores_td))
#
# # Create a color palette from yellow to red
color_palette_degree_td <- colorRampPalette(c("yellow", "red"))(max_degree_value_td + 1)

V(g_90)$ColorDT <- color_palette_degree_td[degree(g_90)]
V(g_00)$ColorDT <- color_palette_degree_td[degree(g_00)]
V(g_10)$ColorDT <- color_palette_degree_td[degree(g_10)]
V(g_20)$ColorDT <- color_palette_degree_td[degree(g_20)]


######Color for DegIn####
# # Find the maximum degree score for each vertex across all networks
max_degree_value_di <- max(unlist(degree_scores_di))
#
# # Create a color palette from yellow to red
color_palette_degree_di <- colorRampPalette(c("yellow", "red"))(max_degree_value_di + 1)

V(g_90)$ColorDI <- color_palette_degree_di[degree(g_90)]
V(g_00)$ColorDI <- color_palette_degree_di[degree(g_00)]
V(g_10)$ColorDI <- color_palette_degree_di[degree(g_10)]
V(g_20)$ColorDI <- color_palette_degree_di[degree(g_20)]
```

```
######Color for DegOut####
# # Find the maximum degree score for each vertex across all networks
max_degree_value_do <- max(unlist(degree_scores_do))
#
# # Create a color palette from yellow to red
color_palette_degree_do <- colorRampPalette(c("yellow", "red"))(max_degree_value_do + 1)

V(g_90)$ColorDO <- color_palette_degree_do[degree(g_90)]
V(g_00)$ColorDO <- color_palette_degree_do[degree(g_00)]
V(g_10)$ColorDO <- color_palette_degree_do[degree(g_10)]
V(g_20)$ColorDO <- color_palette_degree_do[degree(g_20)]



######Color for Hub####

# Create a color palette from yellow to red; using 121 as max value and squaring all
# the hub and auth scores to create greater gradation in the coloring, i.e. make more
# obvious differences in scores via the associated vertex colors.

color_palette_hub <- colorRampPalette(c("yellow", "red"))(121)

V(g_90)$ColorHub <- color_palette_hub[V(g_90)$Hub^2]
V(g_00)$ColorHub <- color_palette_hub[V(g_00)$Hub^2]
V(g_10)$ColorHub <- color_palette_hub[V(g_10)$Hub^2]
V(g_20)$ColorHub <- color_palette_hub[V(g_20)$Hub^2]

######Color for Auth####

# Create a color palette from yellow to red; using 121 as max value and squaring all
```

```
# the hub and auth scores to create greater gradation in the coloring, i.e. make more
# obvious differences in scores via the associated vertex colors.

color_palette_auth <- colorRampPalette(c("yellow", "red"))(121)

V(g_90)$ColorAuth <- color_palette_auth[V(g_90)$Auth^2]
V(g_00)$ColorAuth <- color_palette_auth[V(g_00)$Auth^2]
V(g_10)$ColorAuth <- color_palette_auth[V(g_10)$Auth^2]
V(g_20)$ColorAuth <- color_palette_auth[V(g_20)$Auth^2]

#Adding edge attribute, i.e. thickness based on weight as of migration flow####

######For 1990####
# # Network from EdgeList
EL_Network_90 <- read.csv("1990_edgelist_Per1000.csv", header = T)
#
# # Create a graph from the edgelist
g_EL_90 <- graph_from_data_frame(EL_Network_90, directed = T)
#
# # Scale the edge thickness based on the values
# # You can adjust the scaling factor as needed
E(g_90)$width <- EL_Network_90$Weight


######For 2000####
# Network from EdgeList
EL_Network_00 <- read.csv("2000_edgelist_Per1000.csv", header = T)



# Create a graph from the edgelist
```

```
g_EL_00 <- graph_from_data_frame(EL_Network_00, directed = T)

# Scale the edge thickness based on the values
# You can adjust the scaling factor as needed
E(g_00)$width <- EL_Network_00$Weight


######For 2010####

# Network from EdgeList
EL_Network_10 <- read.csv("2010_edgelist_Per1000.csv", header = T)


# Create a graph from the edgelist
g_EL_10 <- graph_from_data_frame(EL_Network_10, directed = T)

# Scale the edge thickness based on the values
# You can adjust the scaling factor as needed
E(g_10)$width <- EL_Network_10$Weight


######For 2020####
EL_Network_20 <- read.csv("2020_edgelist_Per1000.csv", header = T)


# Create a graph from the edgelist
g_EL_20 <- graph_from_data_frame(EL_Network_20, directed = T)
```

```
# Scale the edge thickness based on the values
E(g_20)$width <- EL_Network_20$Weight


##############################
#Plotting and Saving Jpegs####
##############################

#Set layout for all networks/jpegs for easy comparison

layout_preferred <- layout_with_fr(g_20)

###Degree Total####

jpeg("Plot_USA_SizeDegreeTotal.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))


# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = Degree Total",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$DegT*1.5,
    vertex.color = V(g_90)$ColorDT,
    vertex.frame.color = "lightblue",
```

```
    vertex.label.font = 2)

plot(g_00,
    main = "2000 \n Size & Color = Degree Total",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$DegT*1.5,
    vertex.color = V(g_00)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = Degree Total",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$DegT*1.5,
    vertex.color = V(g_10)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = Degree Total",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$DegT*1.5,
```

```
    vertex.color = V(g_20)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))



######With Edge Attribute####


jpeg("Plot_USA_SizeDegreeTotal_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))

# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = Degree Total",
    edge.arrow.size = 1,
    edge.width = E(g_90)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$DegT,
    vertex.color = V(g_90)$ColorDT,
    vertex.frame.color = "lightblue",
```

```
    vertex.label.font = 2)

plot(g_00,
    main = "2000 \n Size & Color = Degree Total",
    edge.arrow.size = 1,
    edge.width = E(g_00)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$DegT,
    vertex.color = V(g_00)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = Degree Total",
    edge.arrow.size = 1,
    edge.width = E(g_10)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$DegT,
    vertex.color = V(g_10)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = Degree Total",
    edge.arrow.size = 1,
    edge.width = E(g_20)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$DegT,
```

```
    vertex.color = V(g_20)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))


###Degree IN####

jpeg("Plot_USA_SizeDegreeIN.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))


# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = Degree IN",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$DegI*1.5,
    vertex.color = V(g_90)$ColorDI,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)
```

```
plot(g_00,
    main = "2000 \n Size & Color = Degree IN",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$DegI*1.5,
    vertex.color = V(g_00)$ColorDI,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = Degree IN",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$DegI*1.5,
    vertex.color = V(g_10)$ColorDI,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = Degree IN",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$DegI*1.5,
    vertex.color = V(g_20)$ColorDI,
```

```
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()



par(mfrow=c(1,1))



######With Edge Attribute####

jpeg("Plot_USA_SizeDegreeIN_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))

# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = Degree IN",
    edge.arrow.size = 1,
    edge.width = E(g_90)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$DegT,
    vertex.color = V(g_90)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00,
```

```
    main = "2000 \n Size & Color = Degree IN",
    edge.arrow.size = 1,
    edge.width = E(g_00)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$DegT,
    vertex.color = V(g_00)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = Degree IN",
    edge.arrow.size = 1,
    edge.width = E(g_10)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$DegT,
    vertex.color = V(g_10)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = Degree IN",
    edge.arrow.size = 1,
    edge.width = E(g_20)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$DegT,
    vertex.color = V(g_20)$ColorDT,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)
```

```
# Close the JPEG device
dev.off()


par(mfrow=c(1,1))


###Degree Out####

jpeg("Plot_USA_SizeDegreeOUT.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))


# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = Degree OUT",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$DegO*2,
    vertex.color = V(g_90)$ColorDO,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00,
    main = "2000 \n Size & Color = Degree OUT",
```

```
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$DegO*2,
    vertex.color = V(g_00)$ColorDO,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = Degree OUT",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$DegO*2,
    vertex.color = V(g_10)$ColorDO,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = Degree OUT",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$DegO*2,
    vertex.color = V(g_20)$ColorDO,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)
```

```
# Close the JPEG device
dev.off()


par(mfrow=c(1,1))


######With Edge Attribute####

jpeg("Plot_USA_SizeDegreeOUT_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))

# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = Degree OUT",
    edge.arrow.size = 1,
    edge.width = E(g_90)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$DegO,
    vertex.color = V(g_90)$ColorDO,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00,
    main = "2000 \n Size & Color = Degree OUT",
    edge.arrow.size = 1,
    edge.width = E(g_00)$width/4,
```

```
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$DegO,
    vertex.color = V(g_00)$ColorDO,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = Degree OUT",
    edge.arrow.size = 1,
    edge.width = E(g_10)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$DegO,
    vertex.color = V(g_10)$ColorDO,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = Degree OUT",
    edge.arrow.size = 1,
    edge.width = E(g_20)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$DegO,
    vertex.color = V(g_20)$ColorDO,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()
```

```
par(mfrow=c(1,1))


###Hub####

jpeg("Plot_USA_SizeHub.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))


# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = HUB",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$Hub*2,
    vertex.color = V(g_90)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00,
    main = "2000 \n Size & Color = HUB",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
```

```
    layout = layout_preferred,
    vertex.size = V(g_00)$Hub*2,
    vertex.color = V(g_00)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = HUB",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$Hub*2,
    vertex.color = V(g_10)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = HUB",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$Hub*2,
    vertex.color = V(g_20)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()
```

```
par(mfrow=c(1,1))

######With Edge Attribute####

jpeg("Plot_USA_SizeHub_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))



# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = HUB",
    edge.arrow.size = 1,
    edge.width = E(g_90)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$Hub*2,
    vertex.color = V(g_90)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00,
    main = "2000 \n Size & Color = HUB",
    edge.arrow.size = 1,
    edge.width = E(g_00)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$Hub*2,
```

```
    vertex.color = V(g_00)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = HUB",
    edge.arrow.size = 1,
    edge.width = E(g_10)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$Hub*2,
    vertex.color = V(g_10)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = HUB",
    edge.arrow.size = 1,
    edge.width = E(g_20)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$Hub*2,
    vertex.color = V(g_20)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))
```

```
###Authority####

jpeg("Plot_USA_SizeAuth.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))



# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = Auth",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$Auth*2,
    vertex.color = V(g_90)$ColorAuth,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00,
    main = "2000 \n Size & Color = Auth",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$Auth*2,
    vertex.color = V(g_00)$ColorAuth,
```

```
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = Auth",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$Auth*2,
    vertex.color = V(g_10)$ColorAuth,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = Auth",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$Auth*2,
    vertex.color = V(g_20)$ColorAuth,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))
```

```
######With Edge Attribute####


jpeg("Plot_USA_SizeAuth_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))


# Plot each graph
plot(g_90,
    main = "1990 \n Size & Color = Auth",
    edge.arrow.size = 1,
    edge.width = E(g_90)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_90)$Auth*2,
    vertex.color = V(g_90)$ColorAuth,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00,
    main = "2000 \n Size & Color = Auth",
    edge.arrow.size = 1,
    edge.width = E(g_00)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_00)$Auth*2,
    vertex.color = V(g_00)$ColorAuth,
```

```
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10,
    main = "2010 \n Size & Color = Auth",
    edge.arrow.size = 1,
    edge.width = E(g_10)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_10)$Auth*2,
    vertex.color = V(g_10)$ColorAuth,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20,
    main = "2020 \n Size & Color = Auth",
    edge.arrow.size = 1,
    edge.width = E(g_20)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred,
    vertex.size = V(g_20)$Auth*2,
    vertex.color = V(g_20)$ColorAuth,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))
```

```
#######################################################
#######################################################
### WITHOUT THE UNITED STATES INCLUDED #################
#######################################################
#######################################################


#Remove the US from all four networks####

# g_20_NUSA <- delete_vertices(g_20, 17)
# g_10_NUSA <- delete_vertices(g_10, 17)
# g_00_NUSA <- delete_vertices(g_00, 17)
# g_90_NUSA <- delete_vertices(g_90, 17)


#Adding Degree as vertex attribute####

###Degree Total ####
# Assuming you have four graphs g1, g2, g3, g4
graphs_td_NUSA <- list(g_90_NUSA, g_00_NUSA, g_10_NUSA, g_20_NUSA)
#
## Calculate degree scores for each graph
degree_scores_td_NUSA <- lapply(graphs_td_NUSA, degree)


V(g_90_NUSA)$DegT  <- degree_scores_td_NUSA[[1]]
V(g_00_NUSA)$DegT  <- degree_scores_td_NUSA[[2]]
V(g_10_NUSA)$DegT  <- degree_scores_td_NUSA[[3]]
```

```
V(g_20_NUSA)$DegT  <- degree_scores_td_NUSA[[4]]


###Degree IN ####
# Assuming you have four graphs g1, g2, g3, g4
graphs_di_NUSA <- list(g_90_NUSA, g_00_NUSA, g_10_NUSA, g_20_NUSA)
#
## Calculate degree scores for each graph
degree_scores_di_NUSA <- lapply(graphs_di_NUSA, degree, mode = "in")


V(g_90_NUSA)$DegI  <- degree_scores_di_NUSA[[1]]
V(g_00_NUSA)$DegI  <- degree_scores_di_NUSA[[2]]
V(g_10_NUSA)$DegI  <- degree_scores_di_NUSA[[3]]
V(g_20_NUSA)$DegI  <- degree_scores_di_NUSA[[4]]


###Degree OUT ####
# Assuming you have four graphs g1, g2, g3, g4
graphs_do_NUSA <- list(g_90_NUSA, g_00_NUSA, g_10_NUSA, g_20_NUSA)
#
## Calculate degree scores for each graph
degree_scores_do_NUSA <- lapply(graphs_do_NUSA, degree, mode = "out")


V(g_90_NUSA)$DegO  <- degree_scores_do_NUSA[[1]]
V(g_00_NUSA)$DegO  <- degree_scores_do_NUSA[[2]]
V(g_10_NUSA)$DegO  <- degree_scores_do_NUSA[[3]]
V(g_20_NUSA)$DegO  <- degree_scores_do_NUSA[[4]]
```

```
#Creating Hub and Auth Scores####

######Hubs and Auth for 1990####

#Extract and view the hub and authority scores
hub_g_90_NUSA <- hits_scores(g_90_NUSA)$hub
auth_g_90_NUSA <- hits_scores(g_90_NUSA)$authority


######Hubs and Auth for 2000####
hub_g_00_NUSA <- hits_scores(g_00_NUSA)$hub
auth_g_00_NUSA <- hits_scores(g_00_NUSA)$authority


######Hubs and Auth for 2010####
hub_g_10_NUSA <- hits_scores(g_10_NUSA)$hub
auth_g_10_NUSA <- hits_scores(g_10_NUSA)$authority


######Hubs and Auth for 2020####
hub_g_20_NUSA <- hits_scores(g_20_NUSA)$hub
auth_g_20_NUSA <- hits_scores(g_20_NUSA)$authority

###Add Hubs and Auth as Vertex Attributes ####

#Add hubs as vertex attribute
##since scores from 0 to 1, need to times 10 to get reasonable value for attribute use
##and add 1 so that the lowest hub and auth score is not 1, which would affect atrribute use
V(g_90_NUSA)$Hub  <- round(hub_g_90_NUSA*10+1, 0)
```

```
V(g_00_NUSA)$Hub  <- round(hub_g_00_NUSA*10+1, 0)
V(g_10_NUSA)$Hub  <- round(hub_g_10_NUSA*10+1, 0)
V(g_20_NUSA)$Hub  <- round(hub_g_20_NUSA*10+1, 0)

#Add auth as vertex attribute
V(g_90_NUSA)$Auth  <- round(auth_g_90_NUSA*10+1, 0)
V(g_00_NUSA)$Auth  <- round(auth_g_00_NUSA*10+1, 0)
V(g_10_NUSA)$Auth  <- round(auth_g_10_NUSA*10+1, 0)
V(g_20_NUSA)$Auth  <- round(auth_g_20_NUSA*10+1, 0)


#Create color attributes based on DegT, DegI, DEGO, Hub, Auth####


######Color for DegTotal####
# # Find the maximum degree score for each vertex across all networks
max_degree_value_td_NUSA <- max(unlist(degree_scores_td_NUSA))
#
# # Create a color palette from yellow to red
color_palette_degree_td_NUSA <- colorRampPalette(c("yellow", "red"))(max_degree_value_td_NUSA + 1)

V(g_90_NUSA)$ColorDT  <- color_palette_degree_td_NUSA[degree(g_90_NUSA)+1] #Add 1 since Honduras has no edges for 2000 and 2010 and throws an error
V(g_00_NUSA)$ColorDT  <- color_palette_degree_td_NUSA[degree(g_00_NUSA)+1]
V(g_10_NUSA)$ColorDT  <- color_palette_degree_td_NUSA[degree(g_10_NUSA)+1]
V(g_20_NUSA)$ColorDT  <- color_palette_degree_td_NUSA[degree(g_20_NUSA)+1]


######Color for DegIn####
# # Find the maximum degree score for each vertex across all networks
max_degree_value_di_NUSA <- max(unlist(degree_scores_di_NUSA))
#
```

```
# # Create a color palette from yellow to red
color_palette_degree_di_NUSA <- colorRampPalette(c("yellow", "red"))(max_degree_value_di_NUSA + 1)

V(g_90_NUSA)$ColorDI  <- color_palette_degree_di_NUSA[degree(g_90_NUSA)+1]
V(g_00_NUSA)$ColorDI  <- color_palette_degree_di_NUSA[degree(g_00_NUSA)+1]
V(g_10_NUSA)$ColorDI  <- color_palette_degree_di_NUSA[degree(g_10_NUSA)+1]
V(g_20_NUSA)$ColorDI  <- color_palette_degree_di_NUSA[degree(g_20_NUSA)+1]


######Color for DegOut####
# # Find the maximum degree score for each vertex across all networks
max_degree_value_do_NUSA <- max(unlist(degree_scores_do_NUSA))
#
# # Create a color palette from yellow to red
color_palette_degree_do_NUSA <- colorRampPalette(c("yellow", "red"))(max_degree_value_do_NUSA + 1)

V(g_90_NUSA)$ColorDO  <- color_palette_degree_do[degree(g_90_NUSA)+1]
V(g_00_NUSA)$ColorDO  <- color_palette_degree_do[degree(g_00_NUSA)+1]
V(g_10_NUSA)$ColorDO  <- color_palette_degree_do[degree(g_10_NUSA)+1]
V(g_20_NUSA)$ColorDO  <- color_palette_degree_do[degree(g_20_NUSA)+1]



######Color for Hub####

# Create a color palette from yellow to red; using 121 as max value and squaring all
# the hub and auth scores to great greater gradation in the coloring, i.e. make more
# obvious differences in scores via the associated vertex colors.

color_palette_hub_NUSA <- colorRampPalette(c("yellow", "red"))(121)
```

```
V(g_90_NUSA)$ColorHub  <- color_palette_hub_NUSA[V(g_90_NUSA)$Hub ^2]
V(g_00_NUSA)$ColorHub  <- color_palette_hub_NUSA[V(g_00_NUSA)$Hub ^2]
V(g_10_NUSA)$ColorHub  <- color_palette_hub_NUSA[V(g_10_NUSA)$Hub ^2]
V(g_20_NUSA)$ColorHub  <- color_palette_hub_NUSA[V(g_20_NUSA)$Hub ^2]

######Color for Auth####

# Create a color palette from yellow to red; using 121 as max value and squaring all
# the hub and auth scores to great greater gradation in the coloring, i.e. make more
# obvious differences in scores via the associated vertex colors.

color_palette_auth_NUSA <- colorRampPalette(c("yellow", "red"))(121)

V(g_90_NUSA)$ColorAuth  <- color_palette_auth[V(g_90_NUSA)$Auth^2]
V(g_00_NUSA)$ColorAuth  <- color_palette_auth[V(g_00_NUSA)$Auth^2]
V(g_10_NUSA)$ColorAuth  <- color_palette_auth[V(g_10_NUSA)$Auth^2]
V(g_20_NUSA)$ColorAuth  <- color_palette_auth[V(g_20_NUSA)$Auth^2]

##############################
#Plotting and Saving Jpegs####
##############################

#Set layout for all networks/jpegs for easy comparison

layout_preferred_2 <- layout_with_fr(g_20_NUSA)

###Degree Total####

jpeg("Plot_NUSA_SizeDegreeTotal.jpg",
    width = 1500, height = 1500, res = 150)
```

```
# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))



# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = Degree Total",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$DegT *1.5,
    vertex.color = V(g_90_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = Degree Total",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$DegT *1.5,
    vertex.color = V(g_00_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = Degree Total",
    edge.arrow.size = 0.5,
    edge.width = 1,
```

```
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$DegT *1.5,
    vertex.color = V(g_10_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = Degree Total",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$DegT *1.5,
    vertex.color = V(g_20_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))


######With Edge Attribute####


jpeg("Plot_NUSA_SizeDegreeTotal_Thick.jpg",
    width = 1500, height = 1500, res = 150)
```

```
# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))

# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = Degree Total",
    edge.arrow.size = 1,
    edge.width = E(g_90_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$DegT *1.5,
    vertex.color = V(g_90_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = Degree Total",
    edge.arrow.size = 1,
    edge.width = E(g_00_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$DegT *1.5,
    vertex.color = V(g_00_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA\n Size & Color = Degree Total",
    edge.arrow.size = 1,
    edge.width = E(g_10_NUSA)$width/4,
```

```
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$DegT *1.5,
    vertex.color = V(g_10_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = Degree Total",
    edge.arrow.size = 1,
    edge.width = E(g_20_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$DegT *1.5,
    vertex.color = V(g_20_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))


###Degree IN####

jpeg("Plot_NUSA_SizeDegreeIN.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
```

```
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))



# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = Degree IN",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$DegI *1.5,
    vertex.color = V(g_90_NUSA)$ColorDI ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = Degree IN",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$DegI *1.5,
    vertex.color = V(g_00_NUSA)$ColorDI ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = Degree IN",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
```

```
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$DegI *1.5,
    vertex.color = V(g_10_NUSA)$ColorDI ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = Degree IN",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$DegI *1.5,
    vertex.color = V(g_20_NUSA)$ColorDI ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()



par(mfrow=c(1,1))



######With Edge Attribute####

jpeg("Plot_NUSA_SizeDegreeIN_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))
```

```
# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = Degree IN",
    edge.arrow.size = 1,
    edge.width = E(g_90_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$DegT *1.5,
    vertex.color = V(g_90_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = Degree IN",
    edge.arrow.size = 1,
    edge.width = E(g_00_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$DegT *1.5,
    vertex.color = V(g_00_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = Degree IN",
    edge.arrow.size = 1,
    edge.width = E(g_10_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$DegT *1.5,
```

```
    vertex.color = V(g_10_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = Degree IN",
    edge.arrow.size = 1,
    edge.width = E(g_20_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$DegT *1.5,
    vertex.color = V(g_20_NUSA)$ColorDT ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()



par(mfrow=c(1,1))



###Degree Out####

jpeg("Plot_NUSA_SizeDegreeOUT.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))
```

```
# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = Degree OUT",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$DegO *2,
    vertex.color = V(g_90_NUSA)$ColorDO ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = Degree OUT",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$DegO *2,
    vertex.color = V(g_00_NUSA)$ColorDO ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = Degree OUT",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$DegO *2,
    vertex.color = V(g_10_NUSA)$ColorDO ,
```

```
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = Degree OUT",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$DegO *2,
    vertex.color = V(g_20_NUSA)$ColorDO ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()



par(mfrow=c(1,1))



######With Edge Attribute####

jpeg("Plot_NUSA_SizeDegreeOUT_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))

# Plot each graph
plot(g_90_NUSA,
```

```
    main = "1990 No USA \n Size & Color = Degree OUT",
    edge.arrow.size = 1,
    edge.width = E(g_90_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$DegO *2,
    vertex.color = V(g_90_NUSA)$ColorDO ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = Degree OUT",
    edge.arrow.size = 1,
    edge.width = E(g_00_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$DegO *2,
    vertex.color = V(g_00_NUSA)$ColorDO ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = Degree OUT",
    edge.arrow.size = 1,
    edge.width = E(g_10_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$DegO *2,
    vertex.color = V(g_10_NUSA)$ColorDO ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)
```

```
plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = Degree OUT",
    edge.arrow.size = 1,
    edge.width = E(g_20_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$DegO *2,
    vertex.color = V(g_20_NUSA)$ColorDO ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))


###Hub####

jpeg("Plot_NUSA_SizeHub.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))


# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = HUB",
```

```
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$Hub*2,
    vertex.color = V(g_90_NUSA)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = HUB",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$Hub*2,
    vertex.color = V(g_00_NUSA)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = HUB",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$Hub*2,
    vertex.color = V(g_10_NUSA)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)
```

```
plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = HUB",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$Hub*2,
    vertex.color = V(g_20_NUSA)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()



par(mfrow=c(1,1))

######With Edge Attribute####

jpeg("Plot_NUSA_SizeHub_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))



# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = HUB",
    edge.arrow.size = 1,
    edge.width = E(g_90_NUSA)$width/4,
```

```
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$Hub*2,
    vertex.color = V(g_90_NUSA)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = HUB",
    edge.arrow.size = 1,
    edge.width = E(g_00_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$Hub*2,
    vertex.color = V(g_00_NUSA)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = HUB",
    edge.arrow.size = 1,
    edge.width = E(g_10_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$Hub*2,
    vertex.color = V(g_10_NUSA)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = HUB",
```

```
    edge.arrow.size = 1,
    edge.width = E(g_20_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$Hub*2,
    vertex.color = V(g_20_NUSA)$ColorHub,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()



par(mfrow=c(1,1))



###Authority####

jpeg("Plot_NUSA_SizeAuth.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))



# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = Auth",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
```

```
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$Auth *2,
    vertex.color = V(g_90_NUSA)$ColorAuth ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = Auth",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$Auth *2,
    vertex.color = V(g_00_NUSA)$ColorAuth ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = Auth",
    edge.arrow.size = 0.5,
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$Auth *2,
    vertex.color = V(g_10_NUSA)$ColorAuth ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = Auth",
    edge.arrow.size = 0.5,
```

```
    edge.width = 1,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$Auth *2,
    vertex.color = V(g_20_NUSA)$ColorAuth ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()



par(mfrow=c(1,1))



######With Edge Attribute####



jpeg("Plot_NUSA_SizeAuth_Thick.jpg",
    width = 1500, height = 1500, res = 150)

# Set up a 2x2 plotting area
par(mfrow = c(2, 2), mar = c(1, 0, 2, 0))



# Plot each graph
plot(g_90_NUSA,
    main = "1990 No USA \n Size & Color = Auth",
    edge.arrow.size = 1,
    edge.width = E(g_90_NUSA)$width/4,
    vertex.label.cex = .9,
```

```
    layout = layout_preferred_2,
    vertex.size = V(g_90_NUSA)$Auth *2,
    vertex.color = V(g_90_NUSA)$ColorAuth ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_00_NUSA,
    main = "2000 No USA \n Size & Color = Auth",
    edge.arrow.size = 1,
    edge.width = E(g_00_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_00_NUSA)$Auth *2,
    vertex.color = V(g_00_NUSA)$ColorAuth ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_10_NUSA,
    main = "2010 No USA \n Size & Color = Auth",
    edge.arrow.size = 1,
    edge.width = E(g_10_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_10_NUSA)$Auth *2,
    vertex.color = V(g_10_NUSA)$ColorAuth ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

plot(g_20_NUSA,
    main = "2020 No USA \n Size & Color = Auth",
    edge.arrow.size = 1,
```

```
    edge.width = E(g_20_NUSA)$width/4,
    vertex.label.cex = .9,
    layout = layout_preferred_2,
    vertex.size = V(g_20_NUSA)$Auth *2,
    vertex.color = V(g_20_NUSA)$ColorAuth ,
    vertex.frame.color = "lightblue",
    vertex.label.font = 2)

# Close the JPEG device
dev.off()


par(mfrow=c(1,1))


################################
#######Descriptive Stats########
################################

#With US

##Density

edge_density(g_90) #0.126
edge_density(g_00) #0.126
edge_density(g_10) #0.149
edge_density(g_20) #0.187
```

```
##Degree centralization

centr_degree(g_90)$centralization #0.366
centr_degree(g_00)$centralization #0.366
centr_degree(g_10)$centralization #0.370
centr_degree(g_20)$centralization #0.330


##Betweeness Centralization

centr_betw(g_90)$centralization #0.008
centr_betw(g_00)$centralization #0.016
centr_betw(g_10)$centralization #0.029
centr_betw(g_20)$centralization #0.031



#WithOut US#Wg_90ithOut US

##Density

edge_density(g_90_NUSA) #0.085
edge_density(g_00_NUSA) #0.085
edge_density(g_10_NUSA) #0.108
edge_density(g_20_NUSA) #0.150



##Degree centralization
```

```
centr_degree(g_90_NUSA)$centralization #0.066
centr_degree(g_00_NUSA)$centralization #0.159
centr_degree(g_10_NUSA)$centralization #0.166
centr_degree(g_20_NUSA)$centralization #0.183


##Betweenness Centralization

centr_betw(g_90_NUSA)$centralization #0.009
centr_betw(g_00_NUSA)$centralization #0.017
centr_betw(g_10_NUSA)$centralization #0.032
centr_betw(g_20_NUSA)$centralization #0.034




#####################################################
##### Interpreting/analyzing hub vs degree OUT ####
#####################################################

#With US Data

# Calculate hub scores
hub_scores <- hub_score(g_90)$vector

# Order the hub scores from highest to lowest
ordered_indices <- order(hub_scores, decreasing = TRUE)
ordered_hub_scores <- hub_scores[ordered_indices]

# Print the ordered hub scores
```

```
print(ordered_hub_scores)


# Degree OUT
degree_scores <- degree(g_90, mode = "out")

ordered_indices <- order(degree_scores, decreasing = T)
ordered_degree_scores <- degree_scores[ordered_indices]

print(ordered_degree_scores)


#WithOut US

# Calculate hub scores
hub_scores <- hub_score(g_90_NUSA)$vector

# Order the hub scores from highest to lowest
ordered_indices <- order(hub_scores, decreasing = TRUE)
ordered_hub_scores <- hub_scores[ordered_indices]

# Print the ordered hub scores
print(ordered_hub_scores)


# Degree OUT
degree_scores <- degree(g_90_NUSA, mode = "out")
```

```
ordered_indices <- order(degree_scores, decreasing = T)

ordered_degree_scores <- degree_scores[ordered_indices]


print(ordered_degree_scores)




##########################################################################
####

#################### Saving Data Shortcut ####################################

##########################################################################
####



# save(g_90, file="g_90.rda")

# save(g_00, file="g_00.rda")

# save(g_10, file="g_10.rda")

# save(g_20, file="g_20.rda")

#

# save(g_90_NUSA, file="g_90_NUSA.rda")

# save(g_00_NUSA, file="g_00_NUSA.rda")

# save(g_10_NUSA, file="g_10_NUSA.rda")

# save(g_20_NUSA, file="g_20_NUSA.rda")




#BROKERAGE????####################
```

```
install.packages("statnet", dependencies = TRUE)
library(statnet)

install.packages("intergraph")
library(intergraph)

net <- asNetwork(g_20)



brokerage(g_20, cl=network::get.vertex.attribute(net, "name"))
```